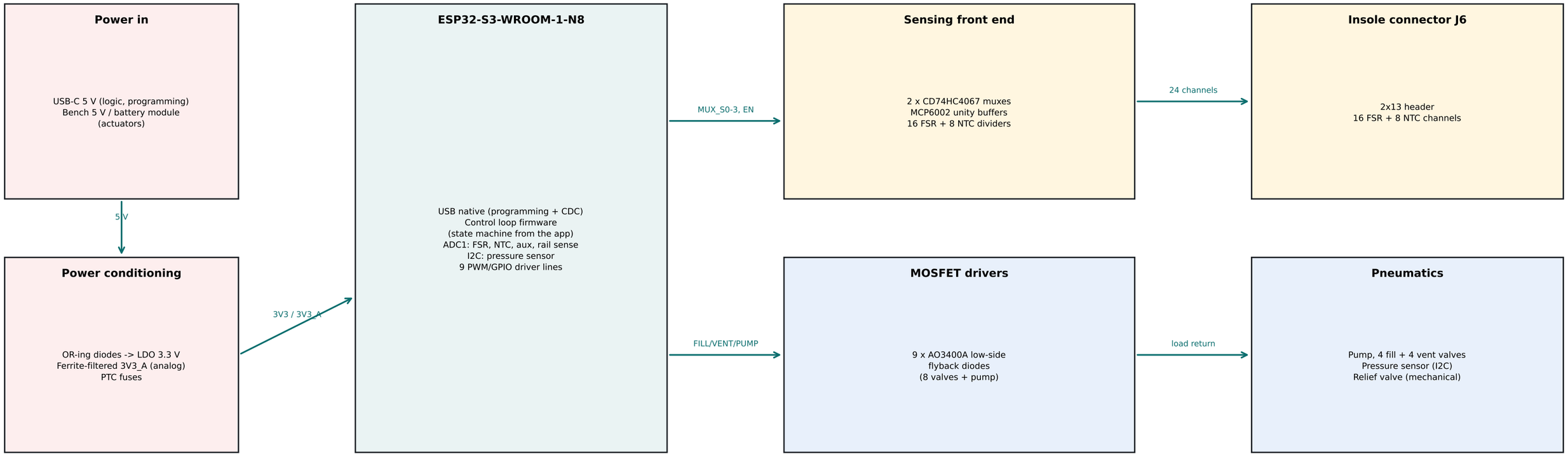


Safety path (independent of firmware)

Normally-closed E-stop switch in series with the actuator rail (J3) -> 2.5 A PTC -> V_ACT.
Actuator power is cut in hardware; the MCU senses V_ACT on IO5 and stays powered.

Safe state: all valves and the pump are normally off (de-energised) when power is lost or the E-stop opens, so every bladder holds its last pressure. Vent valves are normally closed: a powered vent (or the mechanical relief valve) is needed to release pressure. Choose valve types accordingly.

Assumed loads (to confirm with real parts)

MCU module: about 100-350 mA (Wi-Fi bursts) | analog section: <20 mA
Actuator rail sized for 2.5 A (pump about 0.3-1 A, valves about 0.1-0.25 A each, only some energised at once)

GPIO assignment

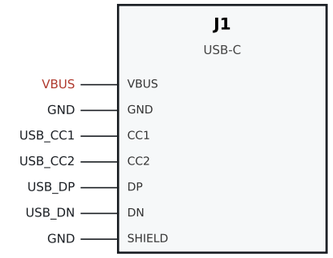
IO0	BOOT	strapping: BOOT button	**IO15**	FILL1	
IO1	FSR_ADC	ADC1_CH0	**IO16**	FILL2	
IO2	NTC_ADC	ADC1_CH1	**IO17**	FILL3	
IO4	AUX_ADC	ADC1_CH3, expansion header	**IO18**	FILL4	
IO5	ACT_SENSE	ADC1_CH4, V_ACT sense	**IO19**	USB_DN	native USB D-
IO6	AUX_IO	expansion header	**IO20**	USB_DP	native USB D+
IO7	LED_STATUS		**IO21**	VENT1	
IO8	I2C_SDA		**IO38**	VENT2	
IO9	I2C_SCL		**IO39**	VENT3	
IO10	MUX_S0		**IO40**	VENT4	
IO11	MUX_S1		**IO41**	PUMP_PWM	
IO12	MUX_S2		**IO42**	BUZZER	
IO13	MUX_S3		**IO43**	UART0_TX	UART0 TX
IO14	MUX_EN_N		**IO44**	UART0_RX	UART0 RX

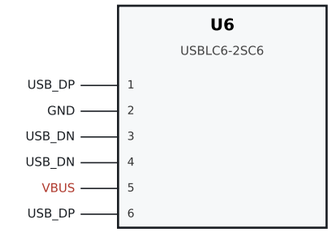
Not used (reserved): IO3, IO45, IO46 (strapping), IO26-IO32 (flash), IO35-IO37 (octal-PSRAM variants), IO47/IO48.
IO39-IO42 are the default pad-JTAG pins; debugging uses the USB Serial/JTAG port instead.

Closed-loop diabetic-foot insole: controller board (ESP32-S3)

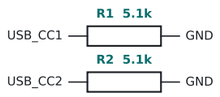
Sheet 1/5: System overview and GPIO map

Rev 0.1 concept, 2026-10-03

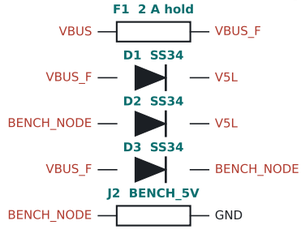




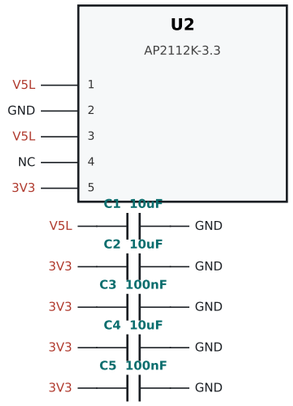
USB-C sink: 5.1k on both CC pins



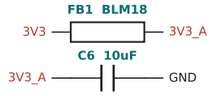
Power input and OR-ing



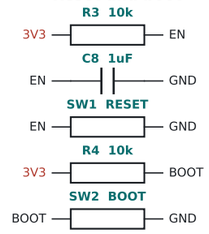
3.3 V LDO and decoupling

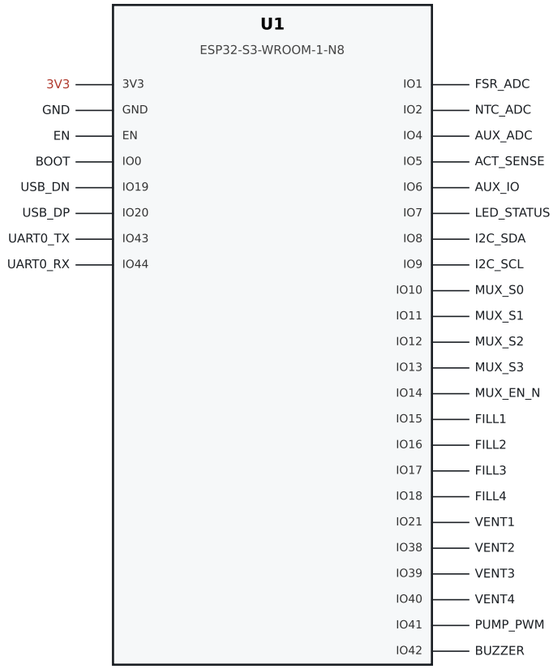


Analog supply filter



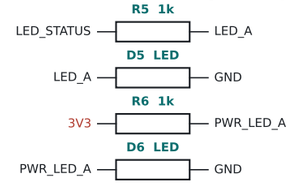
Reset and boot

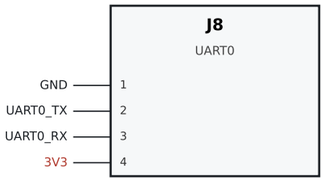


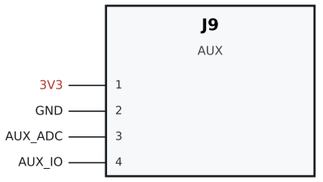


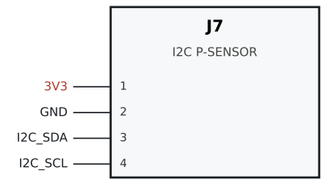
Keep the antenna end clear of copper and metal; no components under the antenna.

Status, debug and expansion

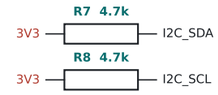








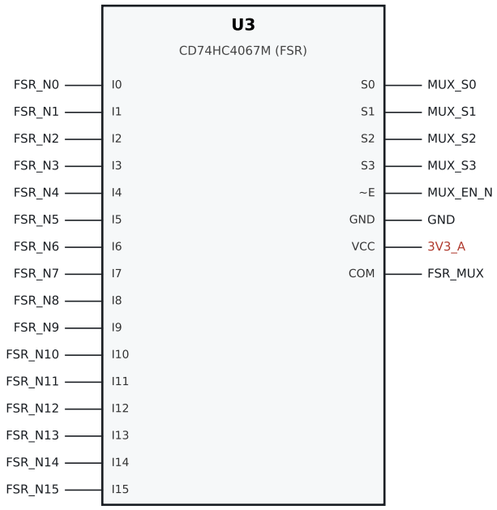
I2C pull-ups

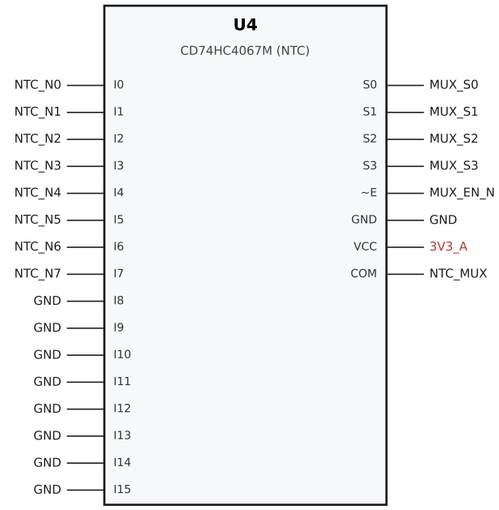


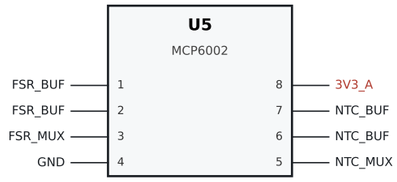
Closed-loop diabetic-foot insole: controller board (ESP32-S3)

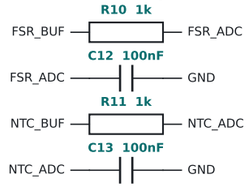
Sheet 2/5: MCU, USB-C, power, indicators, headers

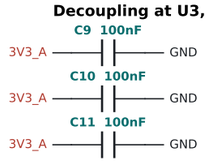
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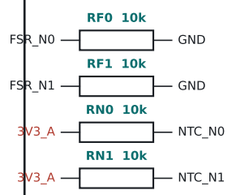






Per-channel dividers

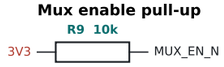
FSR (x16): sensor from 3V3_A (J6 pin 1)
to FSR_Nx; RFx 10k from FSR_Nx to GND.
NTC (x8): RNx 10k 0.1% from 3V3_A to NTC_Nx;
thermistor from NTC_Nx to GND (J6 pin 18).
Unused/unplugged channels read 0 V (FSR)
or near 3V3_A (NTC): no floating inputs.

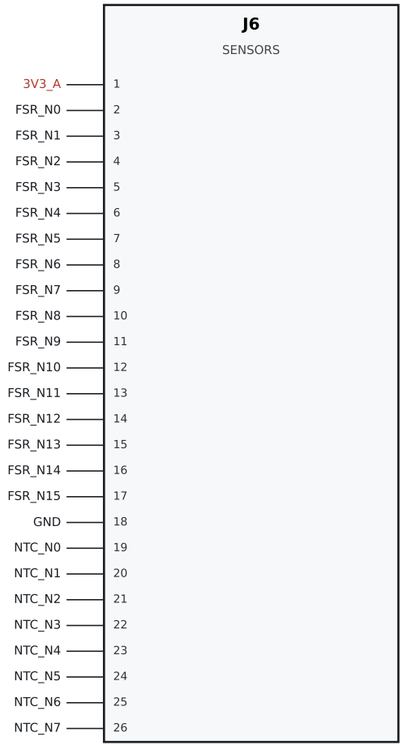


(RF0-RF15 and RN0-RN7 follow the same pattern;
see netlist.csv for all channels.)

Scan: set S0-S3, wait about 0.3 ms, read FSR_ADC (IO1) and NTC_ADC (IO2) together.
U4 inputs I8-I15 are tied to GND. Ron of the mux is
absorbed by the op-amp buffers.

Mux enable pull-up



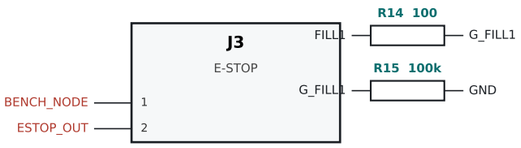


Closed-loop diabetic-foot insole: controller board (ESP32-S3)

Sheet 3/5: Sensor front end: multiplexers, buffers, dividers, insole connector

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Actuator supply and E-stop

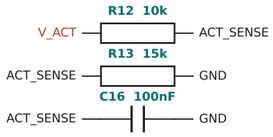


F2 2.5 A hold: ESTOP_OUT — V_ACT

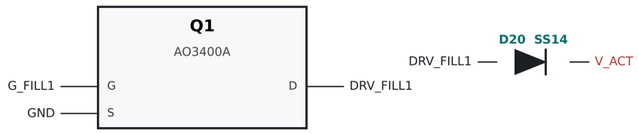
C14 220uF: V_ACT — GND

C15 10uF: V_ACT — GND

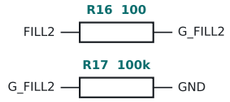
V_ACT sense (to IO5)

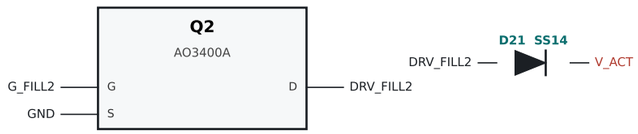


Fill valve 1

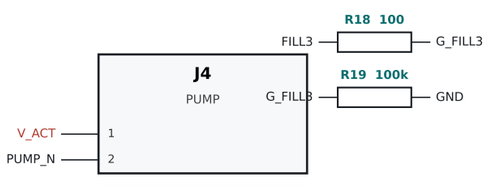


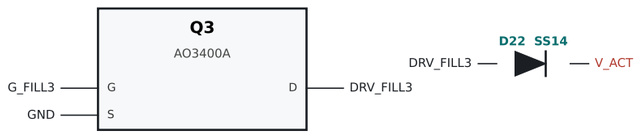
Fill valve 2

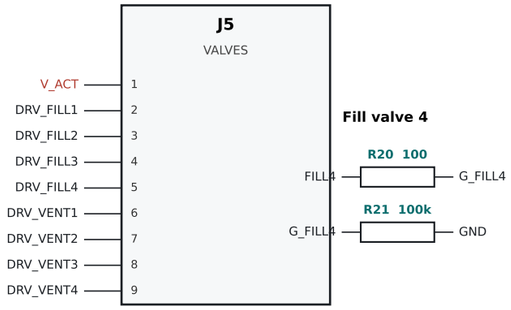




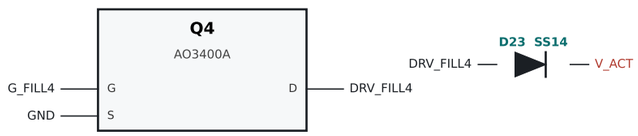
Fill valve 3



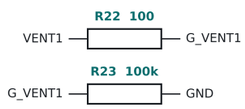


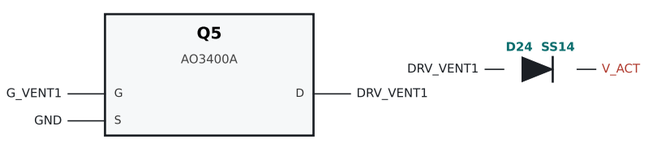


Fill valve 4

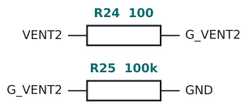


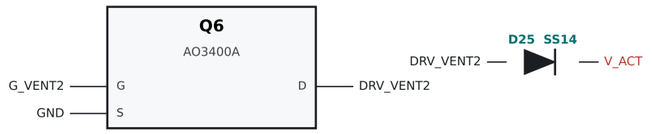
Vent valve 1



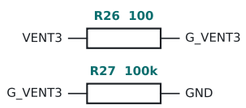


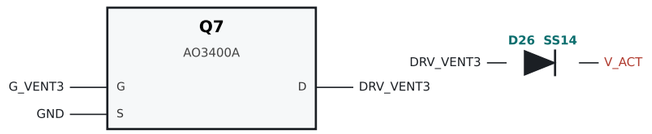
Vent valve 2



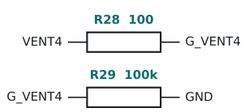


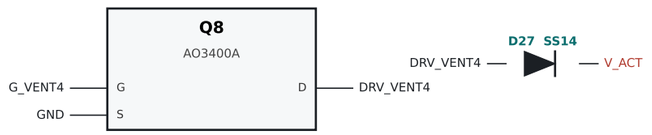
Vent valve 3



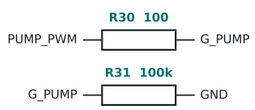


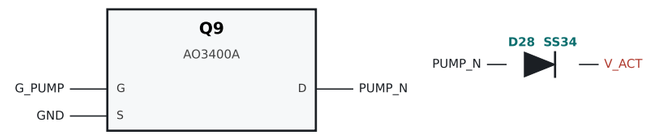
Vent valve 4



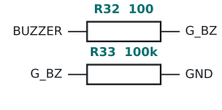


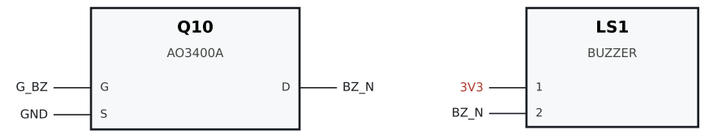
Pump





Buzzer driver





Valve and pump supply V_ACT is common (J5 pin 1, J4 pin 1); each load is switched on its return (low side).

I2C pressure sensor header J7 and pull-ups are on sheet 2.

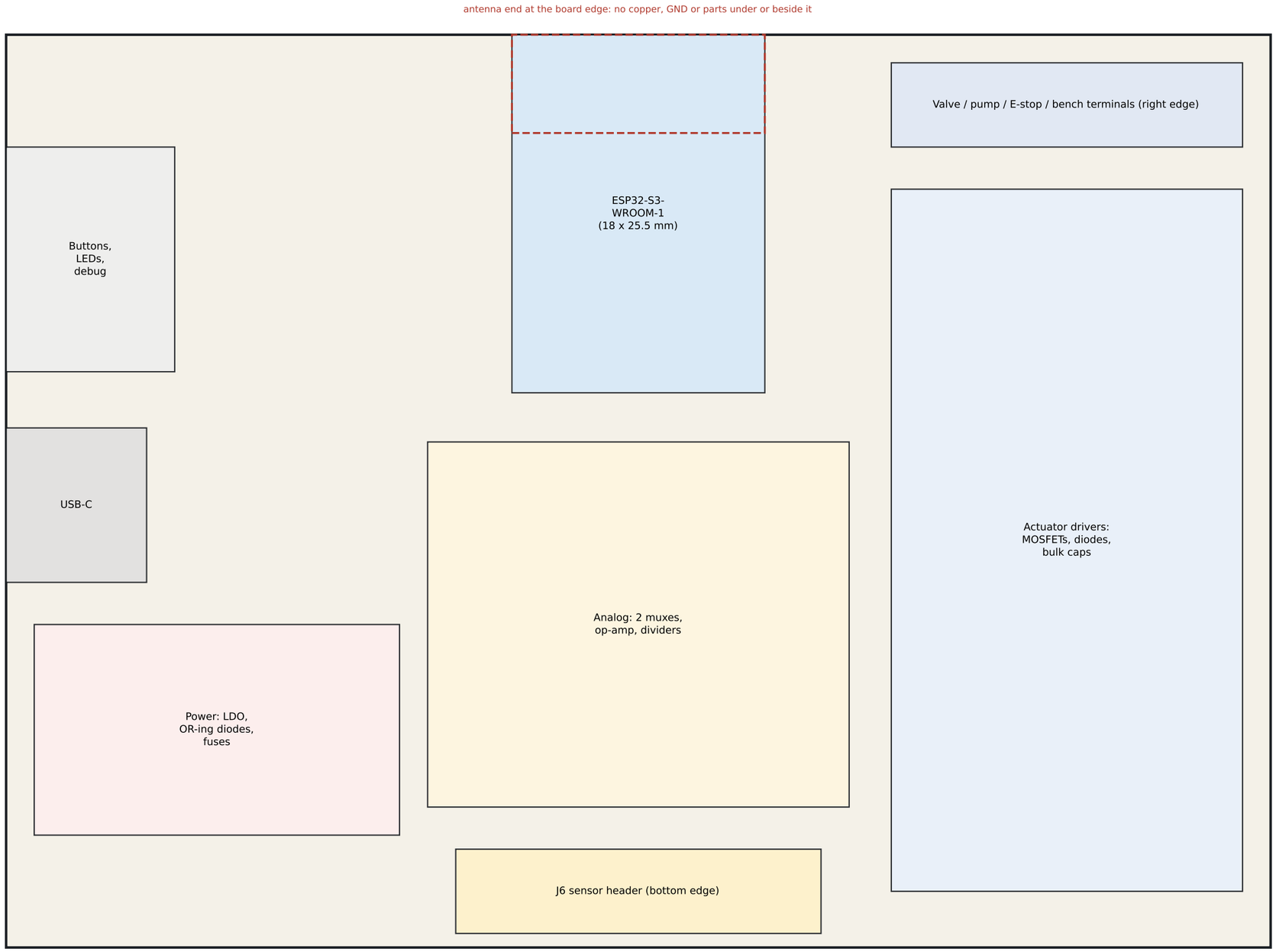
Closed-loop diabetic-foot insole: controller board (ESP32-S3)

Sheet 4/5: Actuator drivers, E-stop, pressure sensor, buzzer

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Placement guidance

1. Antenna at the board edge. 2. USB D+/D- as a short, matched pair with no stubs. 3. Switching loads (pump, valves) on the right, analog on the left.
4. J6 and the ADC filters next to the muxes; keep actuator return currents out from under the analog section. 5. Solid GND plane under the muxes and op-amp.



Board outline about 90 x 65 mm (drawn at 3:1). 4-layer stack recommended: signal / GND / power / signal.

Closed-loop diabetic-foot insole: controller board (ESP32-S3)

Sheet 5/5: Suggested placement (floorplan, not a routed layout)

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